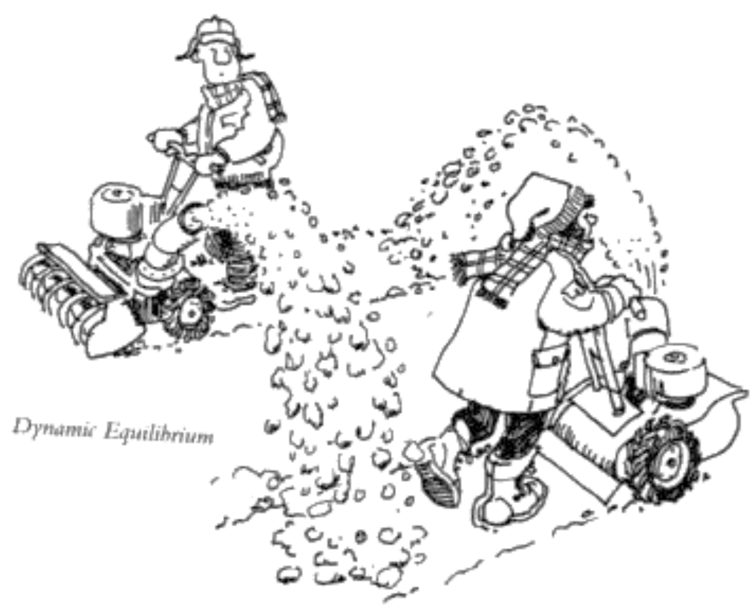
Chapter 15: Chemical Equilibrium



Dr. Wang

Learning Objectives

1. Understand the concept of dynamic equilibrium and relate to chemical kinetics.

2. Express Equilibrium Constant for a balanced chemical reaction, applying the law of mass action.

3. Write Equilibrium Mass Action Expression (Equilibrium Law) for homogeneous reactions and heterogeneous reactions (solid in a liquid)

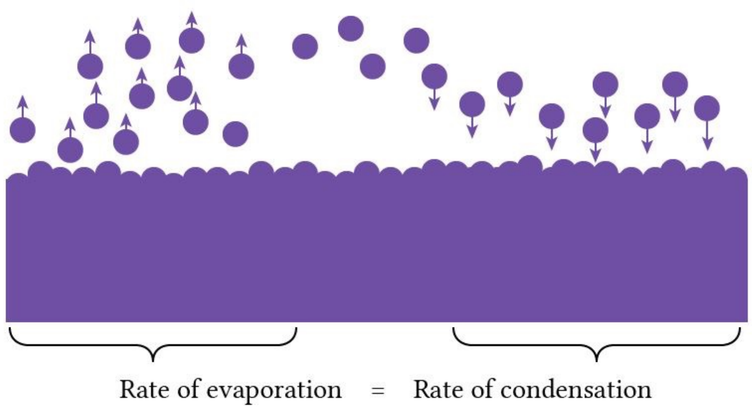
4. Apply the concept of the Equilibrium Constant, K, and determine the extent of product/reactant formation based on the magnitude of K.

5. Calculate equilibrium constant given concentrations

Dynamic Equilibrium

$$\textit{Reactants} \underset{r_{reverse}}{\overset{r_{forward}}{\rightleftarrows}} \textit{Products}$$

- Dynamic Equilibrium exists in chemical reactions that are reversible (can go both forward and backward direction)
- In forward reaction, the reactants reacts to form products.
- In backward reaction, the products reacts to form reactants.



$$H_2O(l) \rightleftharpoons H_2O(g)$$

At the Equilibrium…

- Rate (forward) = Rate (backward)
- Concentrations of the reactants and products are constant).

Dynamic Equilibrium

Starting from $[N_2O_4] \neq 0M$ and $[NO_2] = 0M$:

- In the beginning,

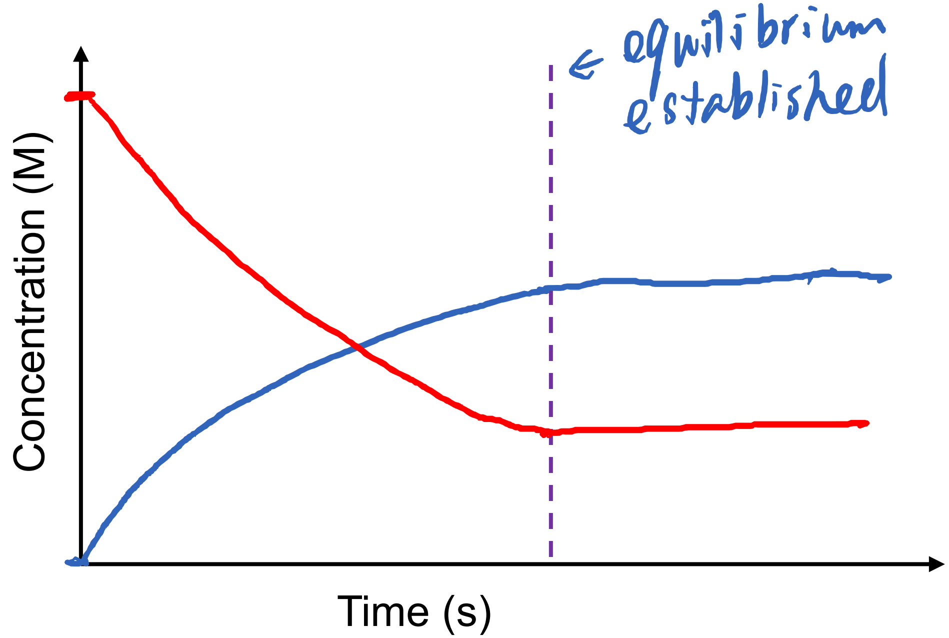
Rate$_{forward}$ = k$_{forward}$ $[N_2O_4]^x$ $>$ Rate$_{backward}$ = k$_{backward}$ $[NO_2]^y$: reaction goes forward

- Before the equilibrium,

$[N_2O_4]$ $\downarrow$ and Rate$_{forward}$ $\downarrow$; $[NO_2]$ $\uparrow$ and Rate$_{backward}$ $\uparrow$

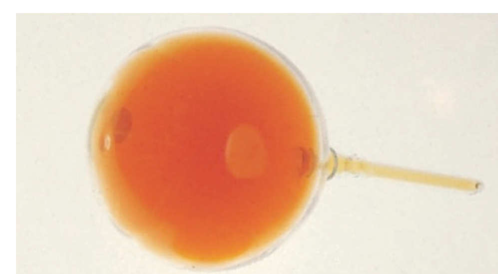
- At the equilibrium,

$[N_2O_4]$ — ; $[NO_2]$ — ; Rate$_{forward}$ $=$ Rate$_{backward}$



$N_2O_4(g) \rightleftharpoons 2NO_2(g)$ $\leftarrow$

colorless brown



Dynamic Equilibrium

Starting from $[N_2O_4] = 0M$ and $[NO_2] \neq 0M$:

- In the beginning,

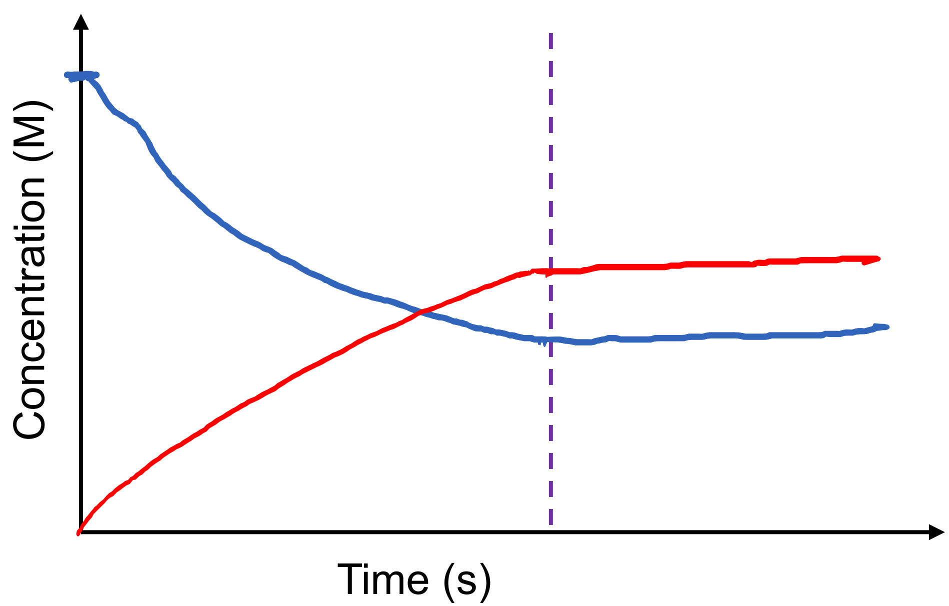
Rate$_{forward}$ $\overset{=0}{=}$ k$_{forward}$ $[N_2O_4]^x$ $<$ Rate$_{backward}$ = k$_{backward}$ $[NO_2]^y$: reaction goes backward

- Before the equilibrium,

$[N_2O_4]$ $\uparrow$ and Rate$_{forward}$ $\uparrow$; $[NO_2]$ $\downarrow$ and Rate$_{backward}$ $\downarrow$

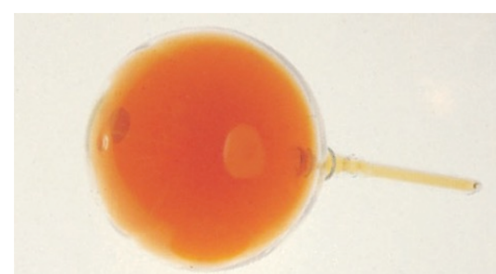
- At the equilibrium,

$[N_2O_4]$ — ; $[NO_2]$ — ; Rate$_{forward}$ $=$ Rate$_{backward}$



$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

colorless brown



Your turn!

Which statement does NOT generally apply to a chemical reaction in dynamic equilibrium?

A. The rates of the forward and reverse reactions are equal.

B. The concentrations of the reactants and products are constant.

C. The concentrations of the reactants and products are equal.

Announcement 4/15

Assignments that are due soon

- Pre-Class Assignment Week 12-M

Chem RePS Session

Wed 2:00 - 3:30 PM at KC 230

2nd Chem RePS or SI worksheets for Unit 3 will be closed on 11:59 PM 4/15

3rd Chem RePS or SI worksheets for Unit 3 will be closed on 10:00 AM 4/17

Office Hour

- Dr. Wang’s: Thu 10-11 AM on Zoom

SI Session

Saige Douglass – Wed 5:30 – 7:30 PM at Hashinger 412

Paige Kwan - Thu 5 - 7 PM at KC134

Equilibrium Constant

aA + bB ⇋ cC + dD

$$K = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

- Equilibrium Constant (K) ONLY depends on temperature.
- When K is …

≪1, $[C]^c[D]^d$ is less than $[A]^a[B]^b$; then backward reaction proceeds close to completion

=1, $[C]^c[D]^d$ is equal to ~~than~~ $[A]^a[B]^b$; then the reaction stop at half way

≫1, $[C]^c[D]^d$ is more than $[A]^a[B]^b$; then forward reaction proceeds close to completion

Table 14.1 The NO_2–N_2O_4 System at 25°C

Initial Concentrations (*M*)		Equilibrium Concentrations (*M*)		Ratio of Concentrations at Equilibrium	
$[NO_2]$	$[N_2O_4]$	$[NO_2]$	$[N_2O_4]$	$\frac{[NO_2]}{[N_2O_4]}$	$\frac{[NO_2]^2}{[N_2O_4]}$
0.000	0.670	0.0547	0.643	0.0851	4.65×10^{-3}
0.0500	0.446	0.0457	0.448	0.102	4.66×10^{-3}
0.0300	0.500	0.0475	0.491	0.0967	4.60×10^{-3}
0.0400	0.600	0.0523	0.594	0.0880	4.60×10^{-3}
0.200	0.000	0.0204	0.0898	0.227	4.63×10^{-3}

0.0547 ≪ 0.643

4.65×10^{-3} ≪ 1

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

≫

$$K = \frac{[NO_2]^2}{[N_2O_4]}$$

Forward / Backward reaction is favored.

Equilibrium Constant

aA + bB ⇋ cC + dD

$$K = \frac{[C]^{c}[D]^{d}}{[A]^{a}[B]^{b}}$$

- Equilibrium Constant (K, K_c) ONLY depends on temperature.
- When K is …

10^{-3}

$\ll 1$, $[C]^c[D]^d$ is much smaller than $[A]^a[B]^b$; then backward reaction proceeds close to completion

$=1$, $[C]^c[D]^d$ is equal to than $[A]^a[B]^b$; then ________ reaction proceeds close to completion stop at half way

$\gg 1$, $[C]^c[D]^d$ is much larger than $[A]^a[B]^b$; then forward reaction proceeds close to completion

10^{3}

- Equilibrium Constant (___) ONLY depends on ________.
- When K is ...

$\ll 1$, $[C]^c[D]^d$ is ____ than $[A]^a[B]^b$; then _____ reaction proceeds close to completion

$=1$, $[C]^c[D]^d$ is ____ than $[A]^a[B]^b$; then _____ reaction proceeds close to completion

$\gg 1$, $[C]^c[D]^d$ is ____ than $[A]^a[B]^b$; then _____ reaction proceeds close to completion

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

$$K = \frac{[NO_2]^2}{N_2O_4}$$

Forward / Backward reaction is favored.

Your turn!

Consider the reaction $N_2O_4\ (g) \rightleftharpoons 2NO_2\ (g)$ where N_2O_4 is a colorless gas and NO_2 is a brown gas. At 5°C the reaction is colorless at equilibrium. This indicates that the equilibrium constant for this reaction at 5°C is

$$\frac{[NO_2]^2}{[N_2O_4]}$$

A. $\gg 1$

B. $=1$

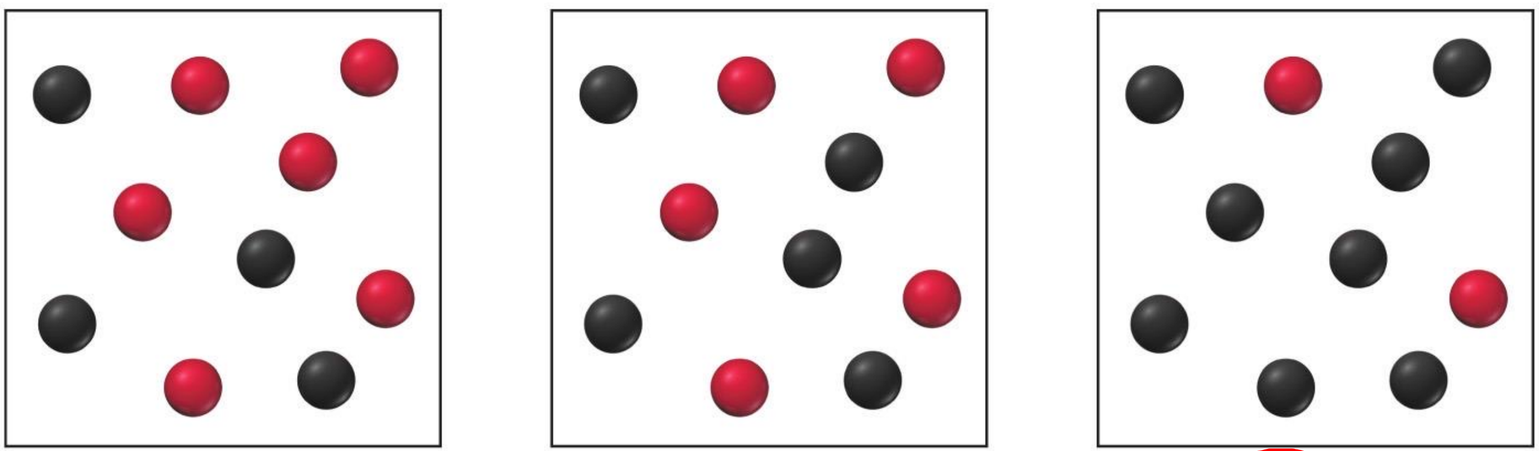
C. $\ll 1$

~~D.~~ $= 0$

~~E.~~ $\ll 0$

Your turn!

Consider the reaction $A(g) \rightleftharpoons B(g)$, The images shown here illustrate equilibrium mixtures of A (red) and B (black) at three different temperatures. At which temperature is the equilibrium constant the largest?



(a) T_1 **(b)** T_2 **(c)** T_3

$$K = \frac{[B]}{[A]}$$

Writing Equilibrium Constant

Kp (partial pressure) X

aA + **bB** ⇋ **cC** + **dD**

$$K = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

Physical states:

- Pure solid (s) and pure liquid (l) are ignored in K. → Kc
- The solute (aq) in aqueous solution, concentration = Mol of solute / L of solution.
- The gas (g) in gaseous mixture, concentration = mol of gas / L of gaseous mixture

$$\underset{X}{AgCl(s)} \rightleftharpoons Ag^+(aq) + Cl^-(aq) \qquad K = \frac{[Ag^+]\times[Cl^-]}{1} = [Ag^+]\times[Cl^-]$$

$$\underset{X}{P_4(s)} + \underset{V}{6Cl_2(g)} \rightleftharpoons \underset{X}{4PCl_3(l)} \qquad K = \frac{1}{[Cl_2]^6}$$

The Unit for K: ignored

Writing Equilibrium Constant

aA + bB ⇋ cC + dD

$$K = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

Physical states:

- Pure solid (s) and pure liquid (l), concentration = 100% ignored in K
- The solute (aq) in aqueous solution, concentration = mol of solute / L of solution.
- The gas (g) in gaseous mixture, concentration = mol of gas / L of gaseous mixture

$$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq) \qquad K = \frac{[Ag^+][Cl^-]}{\cancel{[AgCl]}} = [Ag^+][Cl^-]$$

$$P_4(s) + 6Cl_2(g) \rightleftharpoons 4PCl_3(l) \qquad K = \frac{\cancel{[PCl_3]^4}}{\cancel{[P_4]}[Cl_2]^6} = \frac{1}{[Cl_2]^6}$$

The Unit for K: depends on formula (ignored)

Your turn!

$Ca_3(PO_4)_2(s) \rightleftharpoons 3Ca^{2+}(aq) + 2PO_4^{3-}(aq)$

Consider the reaction: $3Ca^{2+}(aq) + 2PO_4^{3-}(aq) \leftrightharpoons Ca_3(PO_4)_2(s)$. Which equilibrium constant expression is correct for the reversed reaction?

A. $K = \dfrac{[Ca^{2+}]^3[PO_4^{3-}]^2}{[Ca_3(PO_4)_2]}$

B. $K = [Ca^{2+}]^3[PO_4^{3-}]^2$

C. $K = \dfrac{[Ca_3(PO_4)_2]}{[Ca^{2+}]^3[PO_4^{3-}]^2}$

D. $K = \dfrac{1}{[Ca^{2+}]^3[PO_4^{3-}]^2}$ (forward)

Your turn!

Consider the reaction: $3Ca^{2+}(aq) + 2PO_4^{3-}(aq) \leftrightharpoons Ca_3(PO_4)_2(s)$. Which equilibrium constant expression is correct for the reversed reaction?

$Ca_3(PO_4)_2(s) \rightleftharpoons 3Ca^{2+}(aq) + 2PO_4^{3-}(aq)$

A. $K = \dfrac{[Ca^{2+}]^3[PO_4^{3-}]^2}{[Ca_3(PO_4)_2]}$

B. $K = [Ca^{2+}]^3[PO_4^{3-}]^2$

C. $K = \dfrac{[Ca_3(PO_4)_2]}{[Ca^{2+}]^3[PO_4^{3-}]^2}$

D. $K = \dfrac{1}{[Ca^{2+}]^3[PO_4^{3-}]^2}$

Writing Equilibrium Constant

aA + bB ⇋ cC + dD $$K = \frac{[C]^a[D]^d}{[A]^a[B]^b}$$

For the reversed equation:

- **cC + dD ⇋ aA + bB** $K_{reversed} = \frac{1}{K_{forward}}$

The coefficients in equation matters:

- (**aA + bB ⇋ cC + dD**) * ½ $$K' = \frac{[C]^{\frac{c}{2}}[D]^{\frac{d}{2}}}{[A]^{\frac{a}{2}}[B]^{\frac{b}{2}}} = \left(\frac{[C]^c[D]^d}{[A]^a[B]^b}\right)^{\frac{1}{2}} = K^{\frac{1}{2}}$$

- (**aA + bB ⇋ cC + dD**) * 2 $$K'' = K^2$$

$$\frac{[Cl_2]^{10}}{[PCl_5]^4}$$

favor reactants

Adding the equations:

1. $P_4(s) + 6Cl_2(g) \rightleftharpoons 4PCl_3(g)$, $)\times(-1)$ $K_1 = 2.00 \times 10^{19}$
2. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ $)\times 4$ $K_2 = 1.13 \times 10^{-2}$

$4PCl_5(g) \rightleftharpoons P_4(s) + 10Cl_2(g)$, $K_{goal} = ?$

$$K_{goal} = K_1^{-1} \times K_2^{4} = \frac{K_2^4}{K_1} = \frac{(1.13\times10^{-2})^4}{2.00\times10^{19}} = \boxed{8.15\times10^{-28}} \ll 1$$

Writing Equilibrium Constant

$$aA + bB \leftrightharpoons cC + dD \qquad K = \frac{[C]^a[D]^d}{[A]^a[B]^b}$$

For the reversed equation:

- $cC + dD \leftrightharpoons aA + bB$

$$K_{reversed} = \frac{[A]^a[B]^b}{[C]^c[D]^d} = \frac{1}{K}$$

The coefficients in equation matters:

- ($aA + bB \leftrightharpoons cC + dD$) * ½

$$\frac{a}{2}A + \frac{b}{2}B \rightleftharpoons \frac{c}{2}C + \frac{d}{2}D$$

$$K' = \frac{[C]^{\frac{c}{2}}[D]^{\frac{d}{2}}}{[A]^{\frac{a}{2}}[B]^{\frac{b}{2}}} = \left(\frac{[C]^c[D]^d}{[A]^a[B]^b}\right)^{\frac{1}{2}} = K^{\frac{1}{2}}$$

- ($aA + bB \leftrightharpoons cC + dD$) * 2

$$2aA + 2bB \rightleftharpoons 2cC + 2dD$$

$$K'' = \frac{[C]^{2c}[D]^{2d}}{[A]^{2a}[B]^{2b}} = K^2$$

Your turn!

$\times\left(-\frac{1}{2}\right)$

The equilibrium constant for the reaction $\mathrm{A}(g) \rightleftharpoons 2B(g)$ is 0.010. What is the equilibrium constant for the reaction $\mathrm{B}(g) \rightleftharpoons {}^{1}/_{2}\,A(g)$?

$(2.1\times10^{-2})^{-\frac{1}{2}}$

A. 0.0010

$(0.010)^{-\frac{1}{2}}$

B. 1

$= (10^{-2})^{-\frac{1}{2}}$

C. 10

$= 10^{(+2)\times(+\frac{1}{2})}$

D. 100

$= 10$

Your turn!

Determine the value of the equilibrium constant, K_{goal}, for the reaction

$$N_2(g) + O_2(g) + H_2(g) \rightleftharpoons \frac{1}{2}N_2H_4(g) + NO_2(g), \quad K_{\text{goal}} = ?$$

by making use of the following information:

1. $N_2(g) + O_2(g) \rightleftharpoons 2NO(g), \qquad K_1 = 4.10 \times 10^{-31}$
2. $N_2(g) + 2H_2(g) \rightleftharpoons N_2H_4(g), \qquad K_2 = 7.40 \times 10^{-26}$
3. $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g), \qquad K_3 = 6.00 \times 10^{-13}$

1.35×10^{-34}

Your turn!

Determine the value of the equilibrium constant, K_{goal}, for the reaction

$$N_2(g) + O_2(g) + H_2(g) \rightleftharpoons \tfrac{1}{2}N_2H_4(g) + NO_2(g), \quad K_{\text{goal}} = ?$$

by making use of the following information:

$$1.\ (N_2(g) + O_2(g) \rightleftharpoons 2NO(g),) \times \tfrac{1}{2} \quad K_1 = 4.10 \times 10^{-31}$$

$$2.\ (N_2(g) + 2H_2(g) \rightleftharpoons N_2H_4(g)) \times \tfrac{1}{2} \quad K_2 = 7.40 \times 10^{-26}$$

$$3.\ (2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)), \times \tfrac{1}{2} \quad K_3 = 6.00 \times 10^{-13}$$

$$\tfrac{1}{2}N_2(g) + \tfrac{1}{2}O_2(g) \rightleftharpoons \cancel{NO}(g)$$

$$+ \ \tfrac{1}{2}N_2(g) + H_2(g) \rightleftharpoons \tfrac{1}{2}N_2H_4(g)$$

$$+ \ \cancel{NO(g)} + \tfrac{1}{2}O_2(g) \rightleftharpoons NO_2(g)$$

$$N_2(g) + O_2(g) + H_2(g) \rightleftharpoons \tfrac{1}{2}N_2O_4(g) + NO_2(g)$$

$$K = (4.10 \times 10^{-31})^{\frac{1}{2}} \times (7.40 \times 10^{-26})^{\frac{1}{2}} \times (6.00 \times 10^{-13})^{\frac{1}{2}} = 1.35 \times 10^{-34}$$

1.35×10^{-34}

Your turn!

The following equilibrium has been studied at 230 °C:

$$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$$

In one experiment, the concentrations of the reacting species at equilibrium are found to be [NO] = 0.0542 M, [O_2] = 0.127 M, and [NO_2] = 15.5 M.

Calculate the equilibrium constant (K_c) of the reaction at this temperature.

Which direction is favored for this reaction at 230 °C?

6.44×10^5

Your turn!

The following equilibrium has been studied at 230 °C:

$$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$$

In one experiment, the concentrations of the reacting species at equilibrium are found to be $[NO]$ = 0.0542 M, $[O_2]$ = 0.127 M, and $[NO_2]$ = 15.5 M.

Calculate the equilibrium constant (K_c) of the reaction at this temperature.

$$K = \frac{[NO_2]^2}{[NO]^2[O_2]} = \frac{(15.5\,M)^2}{(0.0542\,M)^2 \times (0.127\,M)} = 6.44 \times 10^5$$

Which direction is favored for this reaction at 230 °C? forward

$K >> 1$ the forward reaction is favored and the forward reaction is going completion.

6.44×10^5